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Technical note

A "room-temperature" injection molding/particulate leaching approach for fabrication of biodegradable three-dimensional porous scaffolds

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Abstract

A "room-temperature" injection molding approach combined with particulate leaching (RTIM/PL) has been, for the first time, developed in this work to fabricate three-dimensional porous scaffolds composed of biodegradable polyesters for tissue engineering. In this approach, a "wet" composite of particulate/polymer/solvent was used in processing, and thus the injection was not performed at melting state. Appropriate viscosity and flowability were facilely obtained at a certain solvent content so that the composite was able to be injected into a mould under low pressure at room temperature, which was very beneficial for avoiding thermal degradation of polyesters. As a demonstration, tubular and ear-shaped porous scaffolds were fabricated from biodegradable poly(D,L-lactide-*co*-glycolide) (PLGA) by this technology. Porosities of the resulting scaffolds were as high as 94%. The pores were well interconnected. Besides the well-known characteristics of injection molding to be suitable for automatization of a fabrication process with high repeatability and precision, this RTIM/PL approach is much suitable for tailoring highly porous foams with its advantages flexible for shaping complicated scaffolds, free of thermal degradation and high-pressure machine, etc.
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1. Introduction

Three-dimensional porous scaffolds fabricated from biodegradable polymers have been widely used as temporary extracellular matrices (ECM) and play critical roles in tissue engineering and in situ tissue reconstruction [1–3]. High porosities and interconnected pores of porous scaffolds are required to facilitate cell seeding and adhesion, ECM secretion and eventual tissue regeneration. On the other hand, complicated external shape is also highly desirable from the clinical point of view because the final anatomical shape of a regenerated tissue is basically dependent on the shape of the associated scaffold. Therefore, an ideal scaffolding technique should be able to simultaneously form internal interconnected pores and external anatomical shapes.

A number of fabrication methods have been developed and applied to fabricate porous scaffolds from biodegradable polymers [2–16]. Among them, combination of shaping methods such as casting [12], compression molding [10,13,14] and extrusion [15] with particulate leaching seems one of practical and acceptable selections from the viewpoint of efficiency and cost. Mikos et al. [12] first reported the solvent casting/particulate leaching method to fabricate highly porous membranes and found that particulate leaching was a

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Nomenclature

D_{end}, $D_{0,end}$ outer diameter at the end of the tubular scaffold and the molded composite, respectively
D_{mid}, $D_{0,mid}$ outer diameter in the middle of the tubular scaffold and the molded composite, respectively
L, L_0 length of the tubular scaffold and the molded composite, respectively
η_{susp} viscosity of a suspension system
η_{cont} viscosity of the continuous phase in a suspension system
η_r a relative viscosity equal to the ratio of η_{susp} over η_{cont}
$\phi_{v,disp}$ volume fraction of the particulates (dispersed phase) in the composite
$\phi_{v,sol}$ volume fraction of solvent in the composite
ϕ_{∞} volume fraction of particulates when η_{susp} verges on infinity
$\Phi_{w,partc}$ weight fraction of particulates based on the total weight of particulates and polymer
$\Phi_{v,pore}$ porosity of the resultant scaffold

very successful pore-forming method. As a widely used shaping method in plastics industry, compression molding was, by Thomson et al. [13,14], first combined with particulate leaching to fabricate tissue engineering scaffolds at the processing temperature higher than the glass transition temperature (T_g) of polymers. Recently, we also reported a modified compression molding/particulate leaching method using a novel flexible-rigid combinatory mould to fabricate porous scaffolds with complicated shapes [17]. However, in most of the reported molding approaches, the processing temperatures are higher than the underlying T_g or melting points (T_m), which may lead to thermal degradation of polyesters to a certain degree. It is also not easy to fabricate tubular scaffolds with a small diameter and/or a thin wall due to difficulties to charge mould and to control wall thickness.

Tubular porous scaffolds play important roles in nerve and blood vessel tissue engineering. They can be fabricated by bending and adhesion of porous membranes [18] or casting and freeze drying [19]. Widmer et al. [15] reported an extrusion/particulate leaching process for manufacturing porous conduits. In this approach, temperature higher than 250 °C and pressure higher than 150 MPa in a water press machine are required to extrude the composite of polymer and particulate through the annular space between a nozzle and a rod.

Compared with compression molding, injection molding is characterized by higher reproducibility, automatization and production efficiency. Gomes et al. [20] reported an injection molding/gas foaming approach to prepare biodegradable starch-based polymeric scaffolds with a compact surface and porous cores. Highly porous scaffolds suitable for tissue engineering were, however, not reported.

In this work, we report a novel approach to facilely fabricate highly porous scaffolds with tubular or other complicated shapes under normal conditions. The approach is called room temperature injection molding/particulate leaching (RTIM/PL) by us. According to our approach, a "wet" composite of particulate/polymer/solvent as a suspension system was used as the processing objective. Its viscosity and flowability were elaborately tuned by the solvent content so that it could be injected into a mould under relatively low pressure at room temperature. High porosity and interconnected pore structure were achieved. Differing from the conventional injection molding, this RTIM/PL approach avoids high processing temperature and pressure. It provides a more practical technology to fabricate porous foams composed of biodegradable polymers.

2. Materials and methods

2.1. Materials

Poly(D,L-lactide-*co*-glycolide) (PLGA) with lactide/glycolide molar ratio of 85/15 (PLGA85/15, Purac Co.) and poly(ε-caprolactone) (PCL, Shanghai Hosen Trade Co. Ltd.) were used as scaffold materials. Chloroform was employed as solvent. Sodium chloride particulates (90–180, 180–280 and 280–450 μm) were used as porogens.

2.2. Scaffold fabrication

Porous scaffolds were fabricated via an injection molding at room temperature followed by particulate leaching. A mini-type laboratory-mixing machine (LMM, ATLAS Co.) was used as the injection molder (Fig. 1A). A special designed combinatory mould (Fig. 1B) was used as a necessary accessory to facilitate shaping and mould discharging. For fabrication of a complex-shaped scaffold, the combinatory mould was composed of a flexible inner part and a rigid outer part as previously reported [17]. The rigid outer part was made of metal and employed to make the mould pressure-loadable. The flexible inner part was made of an elastic medical silicone rubber by an in situ cross-linking reaction at room temperature and employed to enhance the shaping ability and to

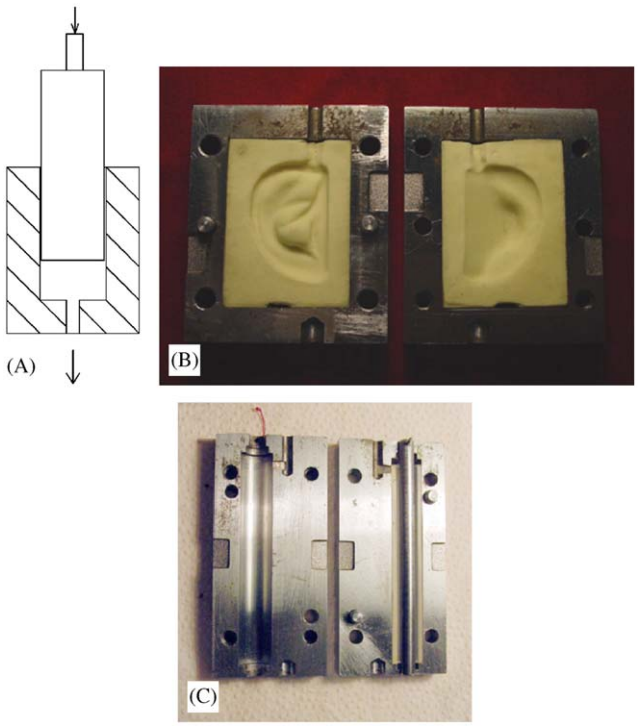

Fig. 1. Diagram of the mini-type injection molder (laboratory-mixing machine, LMM) and the moulds for injection molding. The middle is the mould for fabrication of ear-shaped porous scaffolds and the lower, for fabrication of tubular ones.

facilitate mould release after molding (Fig. 1B). A mould for fabrication of tubular scaffolds is also shown in Fig. 1C.

The polymer was dissolved in an organic solvent to obtain a polymer solution and then sieved particulates were added into the solution to obtain a composite of particulate/polymer/solvent. Such a soft or semisolid paste-like composite was employed as the processing objective in our injection molding process. Low viscosity and good flowability were achieved by elaborately tuning the solvent content in the composite. The "wet" composite was charged into the cylindrical cavity of the injection molder and quickly injected into the mould by pressing the rod under relatively low pressure (by hand) at room temperature.

In the case of low solvent amount, the molded composite was sometimes directly discharged from the mould and kept its shape after de-molding. In order to avoid deformation during drying, a more careful de-molding process is, however, necessary as described as follows. The rigid outer part of the combined mould was first removed. Then the flexible inner part was half opened to make one half of the molded composite exposed to air but the other half still stay in the mould to get support during drying. After air-dried for 24 h and vacuum-dried (<1 mmHg) at room temperature for another 48 h, the composite was completely removed from the flexible inner part for further treatment. In the case of a tubular composite, the rod determining the inner diameter of the scaffold was first drawn out before mould opening. The flexible inner mould part could be removed smoothly after composite drying because of its unique surface character and flexibility.

The dried composite was placed in deionized water to leach out salt particulates. The water was changed every 4 h till no white precipitate was observed as detected by adding silver nitrate solution. Then the obtained porous scaffolds were dried again and stored in a desiccator under vacuum (<1 mmHg) until use.

2.3. Characterization

Porosities of the scaffolds were determined via a modified liquid replacement method [21]. Scanning electron microscopy (SEM, Hitachi S520) was employed to observe pore morphology of the scaffolds.

To determine the shrinkage from the molded composite to the finished scaffold, the length (L) and outer diameters at the end (D_{end}) and in the middle (D_{mid}) of the tubular PLGA scaffolds were measured. L_0, $D_{0,end}$ and $D_{0,mid}$ denote the length, outer diameters at the end and in the middle of the molded tubular composites, which are equal to the corresponding dimensions of the flexible inner mould part. Therefore, the ratio L/L_0, $D/D_{0,mid}$, $D/D_{0,mid}$ were defined as the shrinkage. (See also the nomenclature list for the definitions of most of parameters in this paper) Three samples were measured for each condition.

3. Results and discussion

3.1. Development of injection molding approach at room-temperature

For compression molding [17], the processing objective must be deformable, but its flowability is not necessary. The composite of particulate/polymer with a high particulate content has good ductility at temperature higher than T_g or T_m of the polymer. So, it has been successfully applied in compression molding [13,14,17]. However, for extrusion and injection molding, good flowability is absolutely necessary. To obtain sufficient flowability in extrusion molding, Widmer et al. [15] extruded the composite of particulate/polymer using high temperature (250–270 °C) and pressure (150 MPa).

Can such a composite be used in injection molding process? We first examined its flowability and extrusion or injection behavior using a mini-type LMM. A composite containing 90 wt% particulates with size of 180–280 μm exhibited no flowability at all even among a high temperature range (200–270 °C). Small size and low content of particulates were beneficial for improving its flowability. When a composite contained 80 wt% particulates with size of 90–180 μm, only a little polymer was very slowly extruded while no particulate could be extruded. When the particulate content was decreased to 70%, the flowability was greatly improved and the composite could be extruded at 200 °C. Nevertheless, poor "injectability" was still observed even at higher temperature (240 °C), because much better flowability is

required for injection molding compared with extrusion molding. Further raising temperature or decreasing particulate content may lead to injectability, but cannot be accepted in our study: higher temperature will lead to noticeable thermal degradation of polymer and lower particulate content will result in low porosity and poor pore interconnectivity.

To solve this knotty problem, we tried a "wet" composite of particulate/polymer/solvent instead of a "dry" one of particulate/polymer as the processing objective. It is necessary to mention that a "wet" composite with good deformability has ever been reported by Nam et al. [22] in a gas foaming/particulate leaching processing approach using gas foaming salt as porogen.

Both "dry" composites at melting state and "wet" composites at room temperatrue are typical suspension systems in which solid particulates with volume fraction $\phi_{v,disp}$ are dispersed in a continuous fluid phase. The relation between relative viscosity η_r and $\phi_{v,disp}$ reads [23]

$$\eta_r = \frac{\eta_{susp}}{\eta_{cont}} = \left[1 + \frac{c\phi_{v,disp}}{\phi_\infty - \phi_{v,disp}}\right]^2. \quad (1)$$

Here, η_{susp} and η_{cont} are viscosities of the suspension system and that of the continuous phase, respectively; c is a constant; ϕ_∞ is usually between 60% and 74% related to close packing of the solid particulates [23]. When $\phi_{v,disp}$ is close to ϕ_∞, η_{susp} increases sharply and leads to very poor flowability of the mixture.

In a "dry" NaCl/PCL composite with weight fraction of porogen $\Phi_{w,partc}$ of 90%, $\phi_{v,disp}$ is about 82.7%, which is much higher than 74%. This relation interprets why the composite exhibited very poor flowability and could not be extruded or injected. When $\Phi_{w,partc}$ decreases to 70%, the flowability was found to be greatly improved. But such a low salt fraction might result in poor interconnectivity of the resulting porous scaffolds [12]. For a "wet" composite, the flowability is much improved when a certain content of solvent is employed to adjust $\phi_{v,disp}$ lower than ϕ_∞.

On the other hand, the porosity of a porous scaffold depends mainly on $\Phi_{w,partc}$ (weight fraction of porogen with respect to the total weight of porogen and polymer) for both "dry" and "wet" composites because the solvent in the "wet" composite eventually only influences slightly the porosity of the resulting scaffold due to some small pores generated in solvent evaporation [12]. Therefore, if a "wet" composite is employed, the porosity and flowability may be controlled independently, and the contradiction between them in a "dry" composite is smoothly resolved: high $\Phi_{w,partc}$ ensures high porosity; certain content of solvent leads to low $\phi_{v,disp}$ and thus good flowability.

We used the "wet" composites with a constant $\Phi_{w,partc}$ (90%) to perform injection molding. The results are shown in Table 1. When $\phi_{v,disp}$ is close to or higher than 63%, the injection molding process becomes very difficult. On the other hand, lower $\phi_{v,disp}$ (for example, less than 40%) is unfavorable too, because a shaped composite with high $\phi_{v,sol}$ (volume fraction of solvent in suspention) can not hold its shape after discharged from the mould. Therefore, an appropriate $\phi_{v,disp}$ is in the range of 45–57%.

Different porous scaffolds with tubular and ear-like complex shape were successfully obtained as representively shown in Fig. 2. Such a RTIM approach not only

Table 1
Injection behaviors of different "wet" composites at "room temperature" using a minitype laboratory mixing machine (LMM) and an ear-scaffold mould or a tube-scaffold mould

Fabrication conditions						Results	
Polymer	d (μm)	$\Phi_{w,partc}$ (%)	$\phi_{v,disp}$ (%)	$\phi_{v,sol}$ (%)	Mould	Injection	$\Phi_{v,pore}$ (%)
PCL	90–180	90	45.5	45.7	Tubular	Easy	nd
			52.3	37.6			nd
			57.2	31.8			nd
			63.0	24.9		Difficult	nd
PLGA 85/15	180–280		47.8	43.0		Easy	85.6
			44.6	46.8	Ear-shaped		87.5
			45.2	46.1			90.1
			48.9	41.7			85.7
			49.5	41.0			83.4
			52.0	38.0			82.2
		85	45.3	40.9			76.3
		95	52.8	42.4			94.4
	90–180	90	46.1	45.0			86.9
	280–450		47.5	43.4			85.1

nd: not determined.

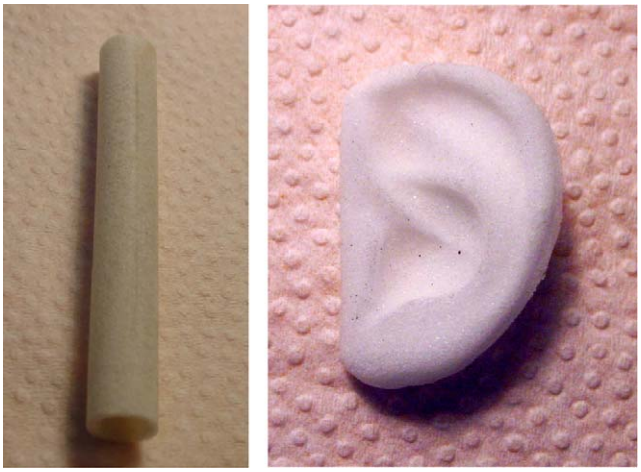

Fig. 2. Photographs of tubular and ear-shaped PLGA85/15 porous scaffolds. (Fabrication conditions: $d = 180–280\,\mu m$; $\Phi_{w,partc} = 90\%$).

ingeniously resolves the contradiction between high porosity and good flowability in injective processing of porous scaffolds, but also avoids material degradation at high temperature and usage of special equipments for high pressure. Furthermore, the RTIM approach is, as is well known, more advantageous for automatization of the fabrication process.

3.2. Pore architecture and porosity

Typical SEM micrographs of a tubular scaffold are shown in Fig. 3. Highly interconnected open pores were uniformly distributed on the inner and outer walls and in the interior of the scaffold. Similar pore morphologies were observed in ear-shaped scaffolds, as shown in Fig. 4.

The measured porosities are given in Table 1. Particualte size and solvent content in the experimental range were found to have relatively little influence on the resulting porosities. The porosity of a porous scaffold fabricated by our RTIM/PL approach is controlled mainly by $\Phi_{w,partc}$.

3.3. Examination of scaffold shrinkage

Injection molding is a nearly net shape molding process and constitutes a very important processing method especially suitable for high repeatability and precision processing. In general, there is little shrinkage from the molded part to the finished one in injection molding of a plastic. The shrinkage depends on the expansion of the material from room temperature to a processing temperature, etc. Since RTIM was employed in this paper, temperature effect can fortunately be neglected. Nevertheless, as the processing objective in our approach is a composite of polymer/particulate/solvent other than polymer itself, the shrinkage behavior

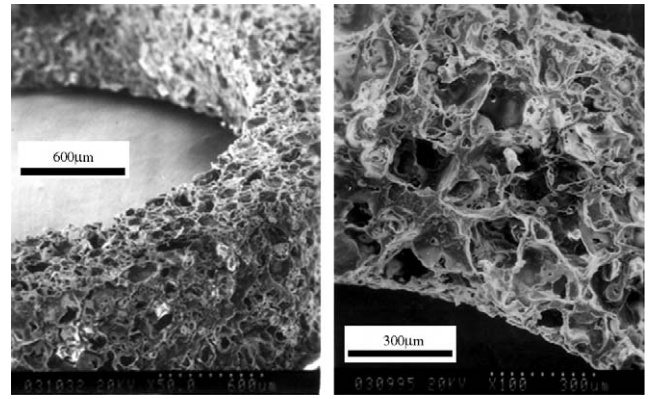

Fig. 3. SEM micrographs of a tubular porous PLGA85/15 scaffold fabricated by RTIM/PL approach. ($d = 180–280\,\mu m$, $\Phi_{w,partc} = 90\%$, $\phi_{v,disp} = 47.8\%$, $\phi_{v,sol} = 43.0\%$).

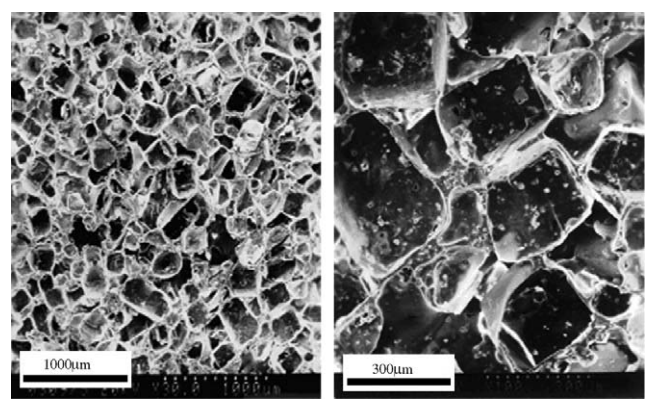

Fig. 4. SEM micrographs of a cross-section of an ear-shaped porous PLGA85/15 scaffold fabricated by RTIM/PL approach. ($d = 280–450\,\mu m$, $\Phi_{w,partc} = 90\%$, $\phi_{v,partc} = 47.5\%$, $\phi_{v,sol} = 43.4\%$).

may be different from the general injection molding process. Therefore, we examined the dimension changes from the molded composites to the resultant porous scaffolds fabricated under different conditions. The results for tubular scaffolds are shown in Table 2.

For fabrication of tubular scaffolds with an outer diameter of 7 mm and a typical wall thickness of 1 mm, the shrinkage is not significant for the examined samples (Table 2). Actually, shrinkage might occur during drying if the molded composite was soft. In our process, the composite was rigidified within few minutes after molded, and the shrinkage became difficult during the following drying.

The small error bars of the shrinkage data demonstrate that the shrinkage is reproducible. In addition, it is uniform throughout the different direction (axial and radial) and position (end and middle) of the scaffold. There is a visual size difference along the length of the conduit foam shown in Fig. 2. It is just due to shooting angles. The tubular scaffold has almost the same

Table 2
Shrinkages from the molded composites to the finished tubular porous scaffolds in the RTIM/PL process under different conditions

Entry	Φ_w (%)	$\Phi_{v,part}$ (%)	L/L_0 (%)	$D/D_{0,end}$ (%)	$D/D_{0,mid}$ (%)
1	90	43.6	99.2±0.4	98.9±0.6	98.5±0.8
2	90	49.9	99.4±0.1	99.5±0.6	99.5±0.1
3	90	52.1	98.8±0.3	99.3±1.6	97.9±0.7
4	90	56.7	98.8±0.2	99.4±0.4	99.3±0.9
5	90	62.1	99.3±0.2	99.5±0.2	99.4±0.5
6	95	49.6	98.9±0.4	97.5±0.4	96.8±1.5
7	85	47.3	99.0±0.5	99.0±0.1	99.0±0.2

diameters at the two ends and in the middle according to quantitative measurements.

4. Conclusions

In this work, highly porous three-dimensional scaffolds composed of biodegradable polyesters were successfully fabricated by an injection molding/particulate leaching approach. The newly developed RTIM/PL approach was conducted at room temperature under relatively low pressure, and consequently avoided thermal degradation of polymers and use of high-pressure equipments. It might be flexible to be applied to different polymers and has been used to fabricate not only complex shaped porous scaffolds but also tubular ones with thin wall and small diameter. The porosity and pore size of the resulting scaffolds are reasonably controlled by ratio of porogen to polymer and particulate size, respectively. The porosity as high as 94% has been achieved. An injection molding approach is more suitable for automatization of a fabrication process. Therefore, we believe that our RTIM/PL approach might be promising for fabrication of porous foams in tissue engineering and in other application fields.

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